

Restoring the black color of alpaca fiber

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Introduction

Based on fiber classification processes within the textile industry, approximately 80% of alpaca fiber corresponds to white fleece, while the remaining proportion is distributed across different colored groups. Black fiber represents less than 1%, according to data from the fiber classification conducted by Inca Tops, which reports that only 0.07-0.4% of black fiber from national production (Burgos *et al.*, 2023). The Inca Group, through its Pacamarca Scientific Station, has undertaken the conservation and promotion of black alpaca by implementing both commercial and productive strategies. These strategies aim not only to produce intense black fleeces but also to conserve other colors in alpacas, using the CIELAB values to guide color selection. Consequently, the conservation of black alpacas requires an analysis of the underlying genetic background. Based on this, the following objectives were established: a) to analyze black color segregation in alpaca population, b) to analyze fiber diameter and fleece weight in black alpacas, and c) to estimate the genetic parameters of colorimetry traits and their correlation with the fiber diameter (FD) and fleece weight (FW) in black alpacas.

Materials & Methods

Animal population, phenotypes, and datasets. The dataset analyzed was obtained from PacoProWeb at the Pacamarca Scientific Station and included only Huacaya alpacas. The black alpacas' color categories included Dark brown or reddish black (RB) and full black (B). Approximately 10 g of fiber were sampled from the mid-side of each animal. Colorimetry values of Lightness (L*), red–green axis (a*), and yellow–blue axis (b*) were measured from 2018 to 2024 following Cruz *et al.* (2021). The pedigree file included 17,184 animals recorded between 1992 and 2025. A total of 5,091 colorimetric records under the CIELAB space, 5,825 fiber diameter records, and 3,467 fleece weight records were analyzed from black alpacas, corresponding to 4,565 alpacas, measured between 2018 and 2024. Animal age ranged from 0.1 to 15.3 years.

Color segregation. To analyze fleece color segregation, the dataset included animals with recorded and measured fleece color under colorimetry. A total of 2,196 birth records of black Huacaya alpacas from the full black alpaca program, born between 2018 and 2024.

Fiber diameter and fleece weight analysis. Samples were evaluated for fiber diameter (FD) and fleece weight (FW) were recorded. A comparison of FW with white alpacas was performed using reference values reported by Cruz *et al.* (2025).

Estimation of genetic parameters for colorimetry. To estimate variance components of colorimetry traits (L*, a*, and b*) in black alpacas and their genetic relationship with the main textile traits (FD and FW), a multi-trait animal model with repeated measures was

applied, using the restricted maximum likelihood (REML) procedure. The model fitted for L*, a*, b*, FD, and GFW was: $y = Xb + Zu + Pp + e$. Fixed effects for all traits included coat color with two levels (RB and B), sex with two levels (male and female), except for fiber diameter where a combined effect of sex $\times$ lactation was considered, with three levels (males, non-lactating females, and lactating females); year of data collection with six levels (2018–2024); and age as linear and quadratic covariates in days. Pedigree recoding and analysis were performed using Endog v4.8. Genetic and phenotypic parameters were estimated using VCE 6.0 softwares.

Results & Discussion

Fleece color segregation. Alpaca fiber color has been grouped into nine basic colors, and mating has been carried out within each color group. However, it was observed that colors other than black can also produce black animals, although in very low percentages. Therefore, the use of the entire pedigree for analysis is justified in this study. In order to restore the black alpaca population, only full black (B) males have been mated with reddish-black (RB) and B female alpacas since 2019 in Pacamarca. A comprehensive analysis of the offspring of the Huacaya black alpaca program from 2018 to 2024 is shown in Figure 1.

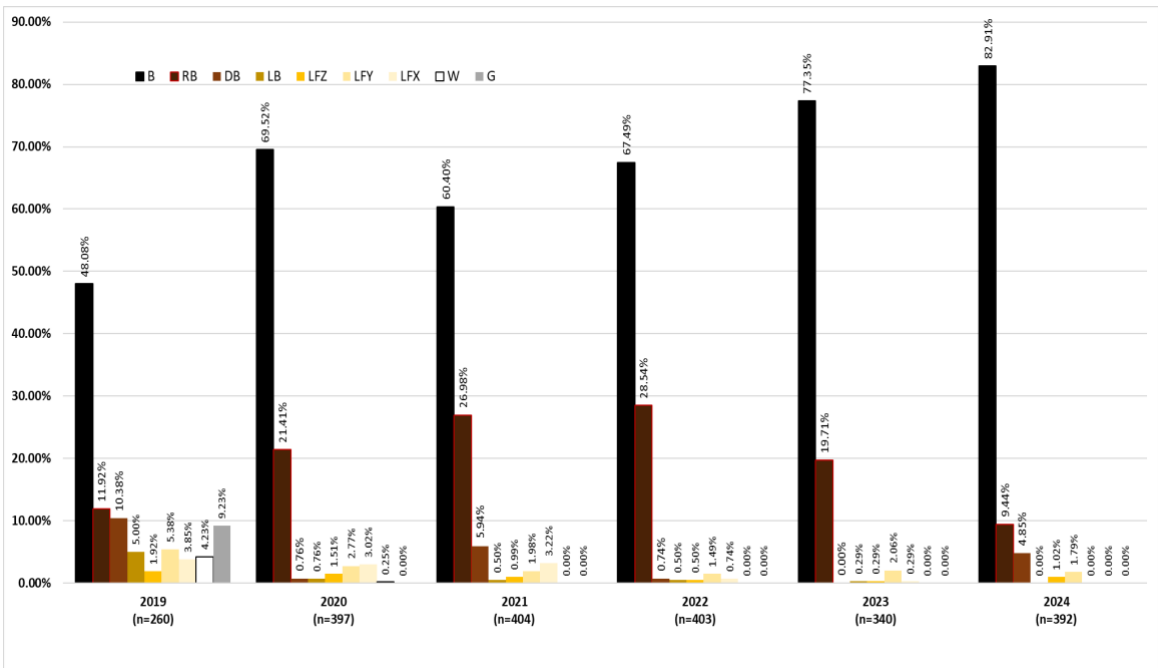


Figure 1. Coat color segregation of offspring from full black alpaca program.

W= White, LFX, LFY y LFZ= light Fawn intensity X, Y and Z, LB= light brown, DB= dark brown, RB= reddish black, B= black and G= all variants of gray, n= number of births

Black Huacaya parents segregated 48.08% of full black offspring in 2019, but this percentage has increased to 82.91% in 2024. On the other hand, it is also observed that the percentage of offspring with light colors such as B, LFX, and LFY has been decreasing, to the point that they no longer segregate in black alpaca parental lines over the years. Particularity, only in the first year did the black alpaca parents segregated the grey color in its various shades (9.23%). The concept of black alpaca does not refer to a single color but rather encompasses both RB and B fiber colors. The inclusion of both within this concept is supported by the segregation data (Figure 1), which shows that RB parents segregated offspring with B and RB coats (38% and 41%, respectively).

Similarly, full black crosses segregated RB and B offspring (16% and 72%, respectively). A mating trial using Peruvian alpacas reported that black × black matings produced only 64% black offspring (Valbonesi *et al.* 2011). Another study reported that black × black crosses yielded 84.7% black progeny in Australia (Paul, 2014), values comparable to those observed in the black alpaca Program.

Fiber diameter and fleece weight in black alpacas. The mean FD of black alpacas (24.59 μm) is lower than that reported by Wuliji *et al.* (2000) for a group of black alpacas in New Zealand (31.60 μm). However, when compared with white Huacayas from the same Pacamarca herd (23.26 μm), the value is slightly higher (Cruz *et al.* 2025). It has already been demonstrated that reducing fiber diameter in alpacas is feasible (Mancisidor *et al.* 2021), suggesting that a staged breeding program could later address this challenge. FW in black alpacas was slightly lower than in white alpacas from the first to the third year of age. From the fourth year of age a significant difference was obtained, with black alpacas producing approximately 700 g less fleece per shearing. The difference in fleece weight performance between white and black alpacas is due to the fact that this selection objective has not been established for black alpacas. However, the strategy of culling during weaning using a subjective fleece density methodology has shown that it can reduce fleece diameter and increase fleece weight in Huacayas alpacas (Cruz *et al.* 2025). The black alpaca program is using a similar strategy to cull less dense animals during weaning. Therefore, it is an opportunity to improve fleece weight in black alpacas.

Heritabilities, genetic and phenotypic correlations of colorimetry and textile traits. Heritabilities, genetic and phenotypic correlations of fiber and colorimetry traits are presented in Table 1. Heritability estimates for CIELAB values were very low (0.01 to 0.04), whereas moderate values were obtained for FD and FW (0.48 and 0.31). Similarly, the ratio of permanent environmental variance was very low to low across all traits (0.01 to 0.19). However, repeatability estimates were low for L* and b*, moderate for a* and FW, and high for FD.

Table 1. Heritability (in the diagonal), genetic correlations (above the diagonal), permanent environmental variances ratio (c^2), and their corresponding standard errors (in brackets), repeatability (R), phenotypic correlations (below the diagonal) for CIELAB values, fiber diameter (FD) and greasy fleece weight (GFW) in Huacaya black alpacas

	L*	a*	b*	FD	FW
L*	0.01 (0.01)	0.66 (0.07)	0.80 (0.07)	-0.55 (0.02)	0.31 (0.03)
a*	0.37	0.04 (0.01)	0.98 (0.02)	-0.99 (0.01)	-0.23 (0.09)
b*	0.43	0.83	0.03 (0.01)	-0.94 (0.04)	-0.10 (0.12)
FD	-0.04	-0.19	-0.28	0.48 (0.02)	0.33 (0.04)
FW	-0.02	-0.08	-0.17	0.43	0.31 (0.03)
c^2	0.01 (0.01)	0.19 (0.02)	0.09 (0.01)	0.12 (0.02)	0.06 (0.02)
R	0.02	0.23	0.12	0.60	0.37

L*= Lightness, a*= axis red - green, b*= axis yellow - blue. All values higher than 0.30 in absolute value are in bold

The estimation of genetic parameters for colorimetry in the black alpaca program revealed heritabilities that were nearly null (<0.04), treating color as a fixed effect has probably caused heritability to drop significantly, using the REML model a significant portion of genetic variability is attributed to this fixed effect. These values were even lower than those reported by Cruz *et al.* (2021), who evaluated the entire population including nine

basic coat colors, and substantially lower than those described in a Spanish population grouped into four color categories (Bartolomé *et al.* 2009). Fiber diameter showed very high and unfavorable genetic correlations with all CIELAB values (–0.55 to –0.99) in black alpacas, indicating that selecting animals for a more intense black coat would strongly increase fiber diameter. Given the magnitude of these correlations, the simultaneous improvement of both traits may be challenging. Similarly, an unfavorable correlation was observed between fleece weight and luminosity, suggesting that animals with deeper black coloration tend to produce lighter fleeces. To address these challenges, a combined selection index is necessary. Or, failing that, abandoning some objectives and giving priority to the production of intense black in this program.

Conclusions

Black alpacas offer a commercial opportunity that can be exploited for the benefit of their conservation. However, selection could be based on colorimetry as a selection criterion, especially L* and b* values. The years spent selecting by coat color are producing very effective results in terms of establishing an intense color. Once this initial objective has been achieved, it could be set aside in order to focus on improving the fineness and weight of the fleece.

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